

Assignment No 6

Smart Grievance Handling System

1) Assign Priority to Issues Based on Severity

Cursor Query:-

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE cursor_assign_priority()
```

```
    -> BEGIN
```

```
    ->     DECLARE done INT DEFAULT 0;
```

```
    ->     DECLARE v_id INT;
```

```
    ->     DECLARE v_severity VARCHAR(10);
```

```
    ->
```

```
    ->     DECLARE cur CURSOR FOR
```

```
    ->         SELECT issue_id, severity FROM Issue;
```

```
    ->
```

```
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
```

```
    ->
```

```
    ->     OPEN cur;
```

```
    ->
```

```
    ->     loop1: LOOP
```

```
    ->         FETCH cur INTO v_id, v_severity;
```

```
    ->
```

```
    ->         IF done THEN
```

```
    ->             LEAVE loop1;
```

```
    ->         END IF;
```

```
    ->
```

```
    ->         IF v_severity = 'High' THEN
```

```
    ->             UPDATE Issue SET priority_score = 90 WHERE issue_id = v_id;
```

```
    ->         ELSEIF v_severity = 'Medium' THEN
```

```
    ->             UPDATE Issue SET priority_score = 60 WHERE issue_id = v_id;
```

```
    ->         ELSE
```

```
    ->             UPDATE Issue SET priority_score = 30 WHERE issue_id = v_id;
```

```
    ->         END IF;
```

```
    ->
```

```
    ->     END LOOP;
```

```
    ->
```

```
    ->     CLOSE cur;
```

```
    -> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

Call:-

```
[mysql> CALL cursor_assign_priority();
```

```
Query OK, 0 rows affected (0.01 sec)
```

Output:-

```
mysql> SELECT issue_id, severity, priority_score  
-> FROM Issue;
```

issue_id	severity	priority_score
1	High	90
2	Medium	60
3	High	90
4	High	90
5	Medium	60
6	High	90
7	Low	30
8	High	90
9	Medium	60
11	High	90
16	High	90

11 rows in set (0.00 sec)

2) Update Status Based on Priority Score

Cursor Query:-

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE cursor_update_status()
```

```
  -> BEGIN
```

```
  ->     DECLARE done INT DEFAULT 0;
```

```
  ->     DECLARE v_id INT;
```

```
  ->     DECLARE v_priority INT;
```

```
  ->
```

```
  ->     DECLARE cur CURSOR FOR
```

```
  ->         SELECT issue_id, priority_score FROM Issue;
```

```
  ->
```

```
  ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
```

```
  ->
```

```
  ->     OPEN cur;
```

```
  ->
```

```
  ->     loop1: LOOP
```

```
  ->         FETCH cur INTO v_id, v_priority;
```

```
  ->
```

```
  ->         IF done THEN
```

```
  ->             LEAVE loop1;
```

```
  ->         END IF;
```

```
  ->
```

```
  ->         IF v_priority >= 80 THEN
```

```
  ->             UPDATE Issue
```

```
  ->             SET status = 'In Progress'
```

```
  ->             WHERE issue_id = v_id;
```

```
  ->         END IF;
```

```
  ->
```

```
  ->     END LOOP;
```

```
  ->
```

```
  ->     CLOSE cur;
```

```
  -> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

Call:-

```
[mysql> CALL cursor_update_status();
```

```
Query OK, 0 rows affected (0.00 sec)
```

Output:-

```
mysql> SELECT
->     issue_id,
->     title,
->     priority_score,
->     status
[ -> FROM Issue;
```

issue_id	title	priority_score	status
1	Pothole near highway	90	In Progress
2	Garbage not collected	60	Open
3	Water leakage	90	In Progress
4	Street light not working	90	In Progress
5	Road damage	60	Open
6	Drainage blockage	90	In Progress
7	Illegal parking	30	Open
8	Tree fallen	90	In Progress
9	Public toilet unclean	60	Open
11	Water shortage	90	In Progress
16	Sewage overflow	90	In Progress

```
11 rows in set (0.00 sec)
```


3) Insert Issue Priority into Log Table

Cursor query:-

```
mysql> CREATE PROCEDURE cursor_insert_log()
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE v_id INT;
->     DECLARE v_priority INT;
->
->     DECLARE cur CURSOR FOR
->         SELECT issue_id, priority_score FROM Issue;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN cur;
->
->     loop1: LOOP
->         FETCH cur INTO v_id, v_priority;
->
->         IF done THEN
->             LEAVE loop1;
->         END IF;
->
->         INSERT INTO Issue_Priority_Log
->         (issue_id, new_priority)
->         VALUES
->         (
->             v_id,
->             v_priority
->         );
->     END LOOP;
->     CLOSE cur;
-> END //
```

Query OK, 0 rows affected (0.00 sec)

Call:-

```
[mysql> CALL cursor_insert_log();
Query OK, 0 rows affected (0.00 sec)
```

Output:-

```
[mysql> SELECT * FROM Issue_Priority_Log;
```

log_id	issue_id	old_priority	new_priority	change_date
1	1	0	90	2026-03-26 18:05:20
2	2	0	60	2026-03-26 18:05:20
3	3	0	90	2026-03-26 18:05:20
4	4	0	30	2026-03-26 18:05:20
5	5	0	60	2026-03-26 18:05:20
6	6	0	90	2026-03-26 18:05:20
7	7	0	30	2026-03-26 18:05:20
8	8	0	90	2026-03-26 18:05:20
9	9	0	60	2026-03-26 18:05:20
12	16	NULL	0	2026-03-29 22:36:33
14	12	0	NULL	2026-03-29 22:40:20
15	4	30	90	2026-03-29 22:43:53
16	11	0	90	2026-04-14 13:56:55
17	16	0	90	2026-04-14 13:56:55
18	7	NULL	30	2026-04-14 14:01:14
19	2	NULL	60	2026-04-14 14:01:14
20	5	NULL	60	2026-04-14 14:01:14
21	9	NULL	60	2026-04-14 14:01:14
22	1	NULL	90	2026-04-14 14:01:14
23	3	NULL	90	2026-04-14 14:01:14
24	4	NULL	90	2026-04-14 14:01:14
25	6	NULL	90	2026-04-14 14:01:14
26	8	NULL	90	2026-04-14 14:01:14
27	11	NULL	90	2026-04-14 14:01:14
28	16	NULL	90	2026-04-14 14:01:14

```
25 rows in set (0.00 sec)
```

4) Delete Low Priority Issues

Cursor query:-

```
mysql> CREATE PROCEDURE cursor_delete_low_priority()  
-> BEGIN  
->     DECLARE done INT DEFAULT 0;  
->     DECLARE v_id INT;  
->     DECLARE v_priority INT;  
->  
->     DECLARE cur CURSOR FOR  
->         SELECT issue_id, priority_score FROM Issue;  
->  
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;  
->  
->     OPEN cur;  
->  
->     loop1: LOOP  
->         FETCH cur INTO v_id, v_priority;  
->  
->         IF done THEN  
->             LEAVE loop1;  
->         END IF;  
->  
->         IF v_priority < 40 THEN  
->             DELETE FROM Issue  
->             WHERE issue_id = v_id;  
->         END IF;  
->  
->     END LOOP;  
->  
->     CLOSE cur;  
-> END //  
Query OK, 0 rows affected (0.00 sec)
```

Call:-

```
[mysql> CALL cursor_delete_low_priority();  
Query OK, 0 rows affected (0.00 sec)
```

Output:-

```
mysql> SELECT
->     issue_id,
->     title,
->     priority_score
[ -> FROM Issue;
```

issue_id	title	priority_score
1	Pothole near highway	90
2	Garbage not collected	60
3	Water leakage	90
4	Street light not working	90
5	Road damage	60
6	Drainage blockage	90
8	Tree fallen	90
9	Public toilet unclean	60
11	Water shortage	90
16	Sewage overflow	90

```
10 rows in set (0.00 sec)
```


5) Count Total Issues Using Cursor

Cursor query:-

```
mysql> CREATE PROCEDURE cursor_count_issues()
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE v_id INT;
->     DECLARE total INT DEFAULT 0;
->
->     DECLARE cur CURSOR FOR
->         SELECT issue_id FROM Issue;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN cur;
->
->     loop1: LOOP
->         FETCH cur INTO v_id;
->
->         IF done THEN
->             LEAVE loop1;
->         END IF;
->
->         SET total = total + 1;
->
->     END LOOP;
->
->     CLOSE cur;
->
->     SELECT total AS total_issues;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

Call & Output:-

```
[mysql> CALL cursor_count_issues();
+-----+
| total_issues |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)
```

6) Find Highest Priority Issue

Cursor Query:-

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE cursor_max_priority()
  -> BEGIN
  ->     DECLARE done INT DEFAULT 0;
  ->     DECLARE v_id INT;
  ->     DECLARE v_priority INT;
  ->     DECLARE max_priority INT DEFAULT 0;
  ->
  ->     DECLARE cur CURSOR FOR
  ->         SELECT issue_id, priority_score FROM Issue;
  ->
  ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
  ->
  ->     OPEN cur;
  ->
  ->     loop1: LOOP
  ->         FETCH cur INTO v_id, v_priority;
  ->
  ->         IF done THEN
  ->             LEAVE loop1;
  ->         END IF;
  ->
  ->         IF v_priority > max_priority THEN
  ->             SET max_priority = v_priority;
  ->         END IF;
  ->
  ->     END LOOP;
  ->
  ->     CLOSE cur;
  ->
  ->     SELECT max_priority;
  -> END //
Query OK, 0 rows affected (0.00 sec)
```

Call & Output:-

```
[mysql> CALL cursor_max_priority();
+-----+
| max_priority |
+-----+
|          90 |
+-----+
1 row in set (0.01 sec)
```

7) Show Issues Area-wise (Filter)

Cursor query:-

```
mysql> CREATE PROCEDURE cursor_akurdi_issues()  
-> BEGIN  
->     DECLARE done INT DEFAULT 0;  
->     DECLARE v_title VARCHAR(255);  
->     DECLARE v_area VARCHAR(100);  
->  
->     DECLARE cur CURSOR FOR  
->         SELECT title, area FROM Issue;  
->  
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;  
->  
->     OPEN cur;  
->  
->     loop1: LOOP  
->         FETCH cur INTO v_title, v_area;  
->  
->         IF done THEN  
->             LEAVE loop1;  
->         END IF;  
->  
->         IF v_area = 'Akurdi' THEN  
->             SELECT v_title, v_area;  
->         END IF;  
->  
->     END LOOP;  
->  
->     CLOSE cur;  
-> END //
```

Query OK, 0 rows affected (0.01 sec)

mysql>

[mysql> DELIMITER ;

Call & Output:-

```
[mysql> CALL cursor_akurdi_issues();  
+-----+-----+  
| v_title          | v_area |  
+-----+-----+  
| Pothole near highway | Akurdi |  
+-----+-----+  
1 row in set (0.00 sec)  
  
+-----+-----+  
| v_title          | v_area |  
+-----+-----+  
| Road damage      | Akurdi |  
+-----+-----+  
1 row in set (0.00 sec)  
  
+-----+-----+  
| v_title          | v_area |  
+-----+-----+  
| Public toilet unclean | Akurdi |  
+-----+-----+  
1 row in set (0.00 sec)  
  
+-----+-----+  
| v_title          | v_area |  
+-----+-----+  
| Water shortage   | Akurdi |  
+-----+-----+  
1 row in set (0.00 sec)  
  
Query OK, 0 rows affected (0.00 sec)
```